Statistical Data Analysis

Lecture 4: Kernel density estimation

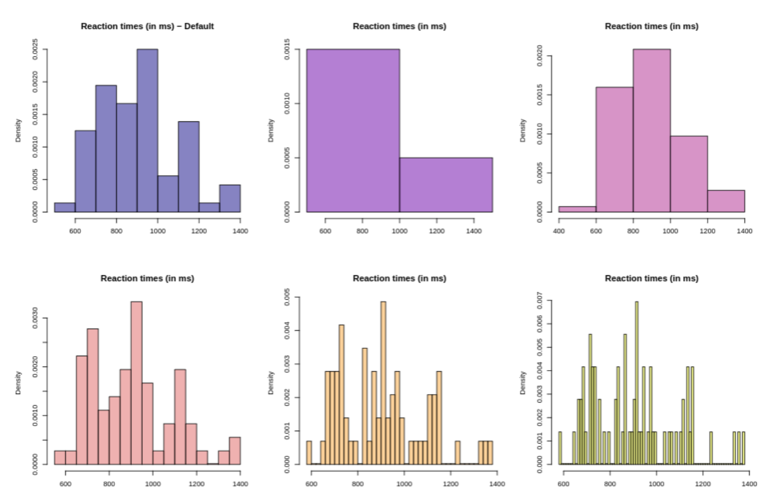
Marc Corstanje

Periods 4 & 5

Where Are We?

Kernel Density Estimation

Histograms as Density Estimators?



Reaction times (in ms) for correct answers given by subject T2 (from `languageR::lexdec`)

Arbitrary bin widths and/or break choices ⇝ visualisation can be misleading

Kernel Density Estimation – Intuition

- X has density f if

$$P(a < X < b) = \int_a^b f(t)\,\mathrm{d}t \quad \text{for all} \quad -\infty < a < b < \infty$$

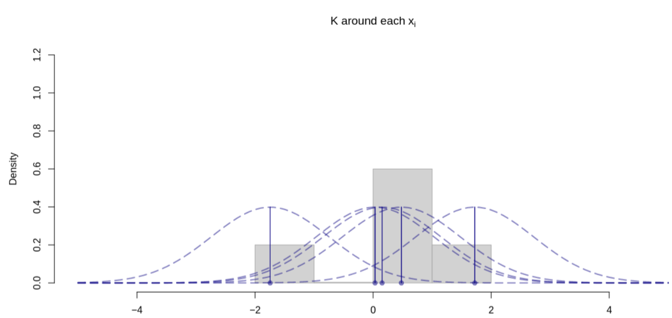
  - Density estimation: reconstruction of true density based on data
  - Sample $x_1, \ldots, x_n$: realizations of i.i.d. random variables $X_1, \ldots, X_n$ from unknown (continuous) distribution F with density f
  - Aim: nonparametric density estimator $t \to \widehat{f}(t)$ (stochastic function)

Kernel Density Estimation – Illustration I

- Kernel K is a given density with expectation 0 and variance 1
- Kernel density estimator:

$$\widehat{f}(t) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{h}K\left(\frac{t-x_i}{h}\right)$$

  - $\widehat{f}$ distributes mass $1/n$ smoothly around each x_i, shaped according to kernel K and with spread of mass specified by bandwidth $h > 0$
  - Intuition: each observed January temperature gives us some confidence that the temperature on a different January day will be close to it

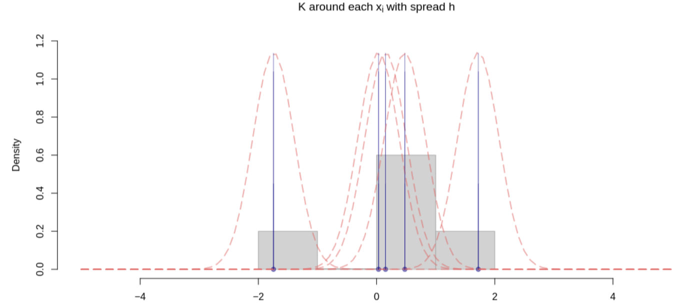


Kernel Density Estimation – Illustration II

- Kernel K is a given density with expectation 0 and variance 1
- Kernel density estimator:

$$\widehat{f}(t) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{h}K\left(\frac{t-x_i}{h}\right)$$

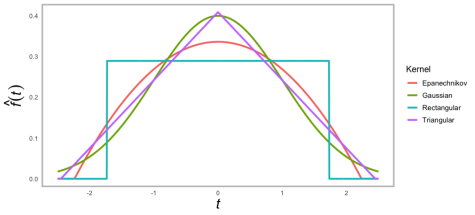
  - $\widehat{f}$ distributes mass $1/n$ smoothly around each x_i, shaped according to kernel K and with spread of mass specified by bandwidth $h > 0$
  - Intuition: a very large bandwidth also ‘covers’ areas away from observations, but hides the underlying data structure (oversmoothing). The opposite is true for a very small bandwidth (undersmoothing)



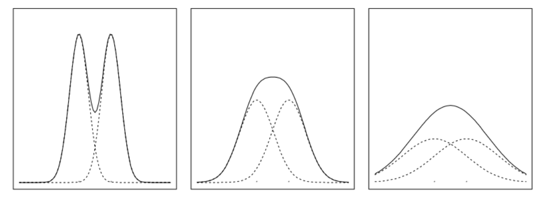
Choice of Kernel and Bandwidth

Choice of Kernel and Bandwidth

- How to choose K and h?
- Kernel choice is less critical:



- Bandwidth choice is more critical:



Gaussian kernel, two observations, different bandwidths ⇝ very different estimators

Choice of Kernel and Bandwidth – MISE

- Objectively optimal choice K and h?
  - Firstly: for which criterion?

Choice of Kernel and Bandwidth – MISE

- Objectively optimal choice K and h?
  - Firstly: for which criterion?
- Possible criterion: choose K and h such that they minimize MISE:

$$\mathrm{MISE}(\widehat{f}) = \int \mathrm{MSE}\left(\widehat{f}(t)\right)\,\mathrm{d}t = \int \underbrace{\mathrm{var}\left(\widehat{f}(t)\right)}_{\text{variance}}\,\mathrm{d}t + \int \underbrace{\left(\mathbb{E}(\widehat{f}(t)) - f(t)\right)^2}_{\text{bias}^2}\,\mathrm{d}t$$

Choice of Kernel and Bandwidth – MISE

- Objectively optimal choice K and h?
  - Firstly: for which criterion?
- Possible criterion: choose K and h such that they minimize MISE:

$$\mathrm{MISE}(\widehat{f}) = \int \mathrm{MSE}\left(\widehat{f}(t)\right)\,\mathrm{d}t = \int \underbrace{\mathrm{var}\left(\widehat{f}(t)\right)}_{\text{variance}}\,\mathrm{d}t + \int \underbrace{\left(\mathbb{E}(\widehat{f}(t)) - f(t)\right)^2}_{\text{bias}^2}\,\mathrm{d}t$$

- It can be shown that

$$\mathrm{MISE}(\widehat{f}) \lesssim \frac{1}{nh}\int K(x)^2\,\mathrm{d}x + \frac{h^4}{4}\int (f''(t))^2\,\mathrm{d}t$$

  - Bound variance integral: $\int \mathrm{var}(\widehat{f}(t))\,\mathrm{d}t \leq \frac{1}{nh}\int K(x)^2\,\mathrm{d}x$
  - Approximate bias integral: $\int (\mathbb{E}(\widehat{f}(t)) - f(t))^2\,\mathrm{d}t \approx \frac{h^4}{4}\int (f''(t))^2\,\mathrm{d}t$

Choice of Kernel and Bandwidth – Minimizers of MISE

- Minimizing the approximate MISE gives optimal bandwidth

$$h_{opt} = \left\{\int K(x)^2 \, \mathrm{d}x\right\}^{1/5} \left\{\int (f''(t))^2 \, \mathrm{d}t\right\}^{-1/5} n^{-1/5}$$

- Inserting h_{opt} in the approximate MISE gives upper bound:

$$\mathrm{MISE}(\widehat{f}) \lesssim \frac{5}{4} n^{-4/5} \left\{\int K(x)^2 \, \mathrm{d}x\right\}^{4/5} \left\{\int (f''(t))^2 \, \mathrm{d}t\right\}^{1/5}$$

  - Optimal kernel minimizes $\int K(x)^2 \, \mathrm{d}x \rightsquigarrow$ Epanechnikov kernel K_e
  - Kernel choice is less critical, unless $\int K(x)^2 \, \mathrm{d}x \gg \int K_e(x)^2 \, \mathrm{d}x$
  - Remark: Gaussian kernel almost as good as Epanechnikov kernel

Choice of Kernel and Bandwidth – Minimizers of MISE

- Recall optimal bandwidth

$$h_{opt} = \left\{ \int K(x)^2 \, \mathrm{d}x \right\}^{1/5} \left\{ \int (f''(t))^2 \, \mathrm{d}t \right\}^{-1/5} n^{-1/5}$$

  - But f'' is unknown (as true density f is unknown)
  - We can only approximate h_{opt} with $\widehat{h}_{opt}$

Choice of Kernel and Bandwidth – Minimizers of MISE

- Recall optimal bandwidth

$$h_{opt} = \left\{ \int K(x)^2 \, dx \right\}^{1/5} \left\{ \int (f''(t))^2 \, dt \right\}^{-1/5} n^{-1/5}$$

  - But f'' is unknown (as true density f is unknown)
  - We can only approximate h_{opt} with $\widehat{h}_{opt}$
- Idea: assume f belongs to parametric class of distributions
  - For example, assume f is Gaussian with mean μ and variance σ^2
  - Then, $\int (f''(t))^2 \, dt \approx 0.212\sigma^{-5}$ and (using also K Gaussian)

$$\widehat{h}_{opt} = (4\pi)^{-1/10} \left(\frac{3}{8} \pi^{-1/2} \right)^{-1/5} \hat{\sigma} \, n^{-1/5} \approx 1.06 \, \hat{\sigma} \, n^{-1/5}$$

  - Unknown σ is estimated with $\hat{\sigma}$
  - Attention: especially if density multimodal/fluctuating, better use

$$\hat{\sigma} = \min \left(\text{sample sd}, \text{sample IQR}/1.34 \right)$$

Cross Validation

ISE & $R(\widehat{f})$

- Another possible criterion: cross-validation (or out-of-sample testing)
  - No further assumptions on data distribution
- Minimize ISE:

$$\mathrm{ISE}(\widehat{f}) = \int (\widehat{f}(t) - f(t))^2 \,\mathrm{d}t = \int \widehat{f}(t)^2 \,\mathrm{d}t - 2\int \widehat{f}(t) f(t) \,\mathrm{d}t + \underbrace{\int f(t)^2 \,\mathrm{d}t}_{\text{unaffected by } h}$$

  - Thus, minimize $R(\widehat{f}) = \int \widehat{f}(t)^2 \,\mathrm{d}t - 2\int \widehat{f}(t) f(t) \,\mathrm{d}t$
  - But $R(\widehat{f})$ depends on unknown f

Cross Validation

- Idea: replace $R(\widehat{f})$ with $\widehat{R}(\widehat{f})$ (independent of f) and minimize it w.r.t h

Cross Validation

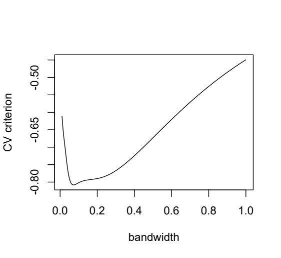
- Idea: replace $R(\widehat{f})$ with $\widehat{R}(\widehat{f})$ (independent of f) and minimize it w.r.t h
- $X_1, \ldots, X_n, Y \overset{i.i.d.}{\sim} F$ with density $f \rightsquigarrow \mathbb{E}(\widehat{f}(Y)|X_1, \ldots, X_n) = \int \widehat{f}(y) f(y)\, \mathrm{d}y$
  - We do not have additional data, so we use cross-validation
  - For each i, estimate f using $x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n \rightsquigarrow \hat{f}_{-i}$

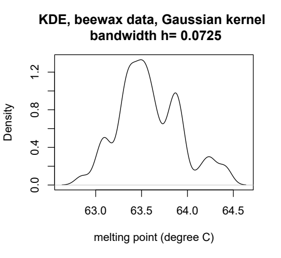
$$\hat{f}_{-i}(t) = \frac{1}{n-1} \sum_{\substack{j=1 \\ j \neq 1}}^{n} \frac{1}{h} K\left(\frac{t - x_j}{h}\right)$$

  - Repeat for each $i = 1, \ldots, n$, then average: $\frac{1}{n} \sum_{i=1}^{n} \widehat{f}_{-i}(x_i) \approx \int \widehat{f}(t) f(t)\, \mathrm{d}t$

Cross Validation

- This leads to $\widehat{R}(\widehat{f}) = \int \widehat{f}(t)^2\,\mathrm{d}t - \frac{2}{n}\sum_{i=1}^{n} \widehat{f}_{-i}(x_i)$
  - If kernel is fixed, minimizing $\widehat{R}(\widehat{f})$ w.r.t. h gives the optimal bandwidth h^*
  - Caveat: $h^* = 0$ is possible $\rightsquigarrow$ always verify that h^* is reasonable





Other Density Estimators

Other Density Estimators

- Problem: kernel density estimates possibly positive in undesirable regions
  - Example: positive random variable, but Gaussian kernel assigns positive mass to all $-\infty < t < \infty$
- Possible solutions
  - Data transformation: set $y_i = \log(x_i) \rightsquigarrow$ derive KDE $\widehat{f}_y$ for y-sample $\rightsquigarrow$ transform back $\widehat{f}_x(t) = \frac{1}{t}\widehat{f}_y(\log(t))$
  - Symmetrization: derive KDE $\hat{f}_S$ based on sample $x_1, -x_1, \dots, x_n, -x_n$ $\rightsquigarrow$ use $t \mapsto 2\hat{f}_S(t)$ for $t > 0$, and 0 otherwise
  - Avoid: just setting $\hat{f}(t)$ to 0 for $t < 0$ and rescale to density

Summary and wrap-up

Summary

- Density estimation requires a kernel and a bandwidth
- Optimal bandwidth and kernel may be found by optimizing the MISE
  - Optimal bandwidth for gaussian data and gaussian kernel $\approx 1.06\hat{\sigma}n^{-1/5}$
  - Optimal kernel: Epanechnikov kernel
- Optimal bandwidth can also be found by optimizing the ISE
  - Use cross-validation
  - Numerically optimize for optimal bandwidth

Wrap-up

- What now?
  - Start working on assignment 3
  - Deadline: 10 March 23:59
  - Material on Bootstrap covered next week